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Anisotropy and tracer-profile assumptions in aperture-based strong-lens dynamics

Conditional inference from supernova distance ratios and unresolved stellar kinematics

Jian Hu[1], Yi Liu[2], Jian-Ping Hu[3], and Zhongmu Li[1]

[1] Institute of Astronomy and Information, Dali University, Dali 671003, People's Republic of China
e-mail: dg1626002@smail.nju.edu.cn
[2] Department of Engineering, Dali University, Dali 671003, People's Republic of China
[3] School of Astronomy and Space Science, Nanjing University, Nanjing 210093, People's Republic of China

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ABSTRACT

Context. Single-aperture stellar kinematics jointly constrain the mass distribution, tracer light profile, and velocity anisotropy. Their coupling affects the interpretation of an inferred anisotropy.
Aims. We quantify how tracer-profile assumptions and data treatment affect common-anisotropy comparisons and inferred mass-density slopes in a compiled galaxy-scale lens sample.
Methods. We analyse 107 lenses using distinct Pantheon+ supernova counterparts, spatial flatness, distance duality, and the full supernova covariance. We compare the individual-slope power-law tracer with Hernquist and Sérsic profiles using measured effective radii. Numerical Jeans solutions, likelihood comparisons, posterior distributions, and controlled recovery experiments quantify the effects of tracer representation.
Results. The power-law fixed-minus-free Bayesian information criterion differences are 6.74 for isotropy and 41.51 for fixed $\beta = 0.18$, compared with −3.89 and −2.26 for Hernquist and −4.67 and −4.30 for Sérsic $n = 4$. The preference also changes with Sérsic index. Removing 12 systems with linearised distance-ratio uncertainty above 20% reduces the power-law isotropy comparison to 2.51. In isotropic Sérsic $n = 4$ and Hernquist simulations, power-law fits yield $\Delta\mathrm{BIC}_0 > 6$ in 60.0% and 50.0% of realisations, respectively, compared with 0.5% and 1.0% for matched-tracer fits.
Conclusions. The preference for a common free anisotropy parameter depends on the tracer representation and sample composition. Controlled recovery identifies tracer mismatch as one mechanism that can generate this preference within the adopted aperture models. Physical interpretation requires joint treatment of photometry, distance uncertainties, and kinematic weighting.

Key words. gravitational lensing: strong – galaxies: elliptical and lenticular, cD – galaxies: kinematics and dynamics – methods: statistical

1. Introduction

Strong gravitational lensing (SGL) connects the projected mass of a foreground galaxy to the geometry between the observer, lens, and background source. Combining lensing observables with stellar kinematics constrains the mass distribution of early-type galaxies (ETGs) and provides information on cosmological distances (Treu 2010; Koopmans et al. 2006; Cao et al. 2015). Time-delay cosmography additionally uses measured delays between variable images to infer distances and the Hubble constant H_0 (Suyu et al. 2017; Wong et al. 2020). Here we study galaxy–galaxy lenses using supernova (SN) distance ratios and aperture-integrated stellar dispersions.

A central difficulty is that lensing and dynamical observables constrain different combinations of the mass profile and stellar orbital structure. Imaging alone admits the mass-sheet degeneracy, while dynamical inference depends on the tracer light distribution, projection, and velocity anisotropy (Falco et al. 1985; Schneider & Sluse 2013; Birrer et al. 2020). Increasing the sample size reduces statistical errors but does not automatically control errors from an inadequate model. Potential biases must therefore be identified, constrained with additional data, and included in the uncertainty budget. Stellar anisotropy is not intrinsically a systematic error: the relevant issue is whether the assumed anisotropy model and its uncertainty are adequate for the data.

Current analyses do not generally fix all lenses to one universal anisotropy value. The TDCOSMO collaboration's 2025 analysis uses a population treatment of spatially constant anisotropy, with uncertainty in the population mean and scatter (TDCOSMO Collaboration 2025). Spatial constancy within a galaxy is distinct from fixing the same known value across galaxies. Forés-Toribio et al. (2026) examine anisotropy alongside point-spread-function (PSF), light-profile, and kinematic-measurement effects. Using simulated galaxies and spatially resolved kinematics, Verma & Minor (2026) find that the recovery of H_0 depends on the adopted anisotropy family and on whether lensing and dynamics are modelled jointly. These studies emphasise the interaction between anisotropy and other modelling choices, rather than the inevitability of an anisotropy-induced bias.

Resolved spectroscopy is also advancing rapidly. The James Webb Space Telescope (JWST) Near-Infrared Spec-

trograph (NIRSpec) integral field unit (IFU) has provided two-dimensional stellar kinematics for RXJ1131−1231 (Shajib et al. 2026). Knabel et al. (2026) extend such measurements to HE0435−1223, PG1115+080, and WFI2033−4723, while Mozumdar et al. (2026) present resolved measurements for B1608+656 and SDSSJ1206+4332, including rotation and neighbouring deflectors. These observations supply radial and spatial information that a single aperture dispersion cannot provide. Our analysis quantifies modelling sensitivity in the complementary regime of compiled aperture-integrated measurements.

We use the individual photometric slopes and lens compilation of Chen et al. (2019), together with supernova-based distance ratios (Hu et al. 2023; Hu 2023; Geng et al. 2025), to quantify the sensitivity of a common-anisotropy preference to the tracer profile and compiled-data approximations. Isotropy and $\beta = 0.18$ are reference models; the historical 0.18 ± 0.13 prescription (Gerhard et al. 2001; Chen et al. 2019) is not a universal description of contemporary lens analyses. Our comparison also separates a fixed value from a probability distribution over values.

Recent work shows how anisotropy interacts with light-profile modelling and kinematic weighting (Forés-Toribio et al. 2026; Verma & Minor 2026). We quantify these interactions in a different observational regime: a catalogue of unresolved aperture dispersions with supernova-derived distance inputs. We compare tracer families on the same matched catalogue and covariance, measure the resulting changes in anisotropy and density-slope inference, and use controlled recovery experiments to assess whether tracer mismatch can generate an effective anisotropy preference. The photometric information entering each tracer family is specified in Sect. 2.4; its implications are discussed in Sect. 3.6.

Section 2 defines the data and forward models. Section 3 presents the likelihood and posterior comparisons, recovery experiments, and sensitivities. Section 4 summarises their interpretation. The appendices document analytic normalisation, covariance propagation, numerical verification, and reproducibility.

2. Data and forward modelling

2.1. Lens sample and survey provenance

Our parent catalogue consists of 130 lenses from the compilation of Chen et al. (2019) for which individual photometric slopes were provided by Y. Chen. We use the subset with the required photometric and kinematic measurements. The observations originate from the Sloan Lens Advanced Camera for Surveys (SLACS) Survey, SLACS for the Masses (S4TM), the Baryon Oscillation Spectroscopic Survey (BOSS) Emission-Line Lens Survey (BELLS), and its Galaxy–Lyα Emitter Systems extension (BELLS GALLERY). Table 1 lists the parent and matched counts with the original survey references. SLACS and S4TM dominate the matched sample, which retains only one BELLS GALLERY lens.

The lens observables are the lens and source redshifts (z_l, z_s), Einstein radius θ_E, aperture radius $\theta_{\rm ap}$, effective light radius $\theta_{\rm eff}$, aperture velocity dispersion $\sigma_{\rm ap}$ and its quoted statistical uncertainty, and the individual luminosity slope δ_i. We use the catalogue measurements directly. The photometric measurements use Hubble Space Telescope (HST) F814W imaging for SLACS, S4TM, and BELLS, and F606W imaging for BELLS GALLERY, as described by Chen et al. (2019).

We use the individual aperture and effective radii listed in the catalogue. We interpret $\theta_{\rm ap}$ as a circular aperture radius and predict the second moment within that aperture. The listed dispersions agree with Table A1 of Chen et al. (2019), before that paper's transformation to $R_e/2$. Direct modelling of the original aperture therefore avoids applying the transformation twice. The heterogeneous spectroscopic extraction and light weighting remain relevant to the sensitivity tests.

2.2. Supernova distance ratios and matching

We use the corrected apparent magnitudes m_b and full statistical-plus-systematic covariance from the official Pantheon+ release (Brout et al. 2022; Scolnic et al. 2022). We match supernovae to the lens and source redshifts using $z_{\rm HD}$ and a fractional comoving-distance tolerance of 0.05 in a flat fiducial background with $\Omega_m = 0.30$. The two-stage mixed-integer linear programme first maximises the number of complete systems and then minimises squared fractional mismatch. Each identifier in the release's CID field is used once globally. A catalogue-order tie break selects a reproducible assignment. The primary sample contains 107 lenses and 214 distinct CIDs. We extract all covariance blocks using the original supernova row indices. The matching metric and fiducial background specify the assignment, while the dynamical fit uses the resulting supernova distance ratios.

For supernovae at the target lens and source redshifts, spatial flatness and distance duality give

$$R \equiv \frac{D_{ls}}{D_s} = 1 - \frac{\chi_l}{\chi_s} = 1 - 10^{0.2(m_{b,l} - m_{b,s})} \frac{1+z_s}{1+z_l}. \tag{1}$$

The common standardised absolute magnitude and an overall distance scale cancel. Differential calibration and redshift-dependent standardisation remain relevant. In practice, the matched supernova redshifts differ from the target redshifts, while the explicit factors in Eq. (1) use the target values. The resulting $R_{\rm SN}$ is therefore a finite-matching approximation. We assess finite matching through alternative tolerances, an input-order control, and model-assisted redshift corrections (Sect. 3.4).

2.3. Common mass model and anisotropy

The baseline two-coefficient (P2) mass model is spherical with $\rho(r) \propto r^{-\gamma}$ and

$$\gamma_i = \gamma_0 + \gamma_z z_{l,i}. \tag{2}$$

The intercept γ_0 gives the extrapolated zero-redshift value of the sample relation, distinct from a radial gradient within a lens. We also report $\gamma_p = \gamma_0 + z_p\gamma_z$ at the sample median $z_p = 0.1959$. This is a transformation of each joint posterior sample, without changing the original prior.

The constant anisotropy is $\beta \equiv \beta_{\rm ani} = 1 - (\sigma_\theta^2 + \sigma_\phi^2)/(2\sigma_r^2)$, shared by the sample. The hypotheses are a free common β, fixed isotropy, and fixed $\beta = 0.18$. Shared spatial constancy is a restrictive model, distinct from a population distribution with intrinsic anisotropy scatter.

2.4. Power-law and curved tracers

For the power-law tracer, $\nu_i(r) \propto r^{-\delta_i}$. The individual slopes were obtained by fitting a PSF-convolved projected profile $I(R) \propto R^{1-\delta_i}$ within $\theta_{\rm eff}/2$ (Chen et al. 2019). In 92 of our 107 lenses, the spectroscopic aperture extends beyond this photometric fitting radius: the median aperture-to-fit-radius ratio is 1.53 and

Table 1. Provenance of the parent catalogue and the baseline matched sample. Counts refer to this analysis, not to the total size of each original survey.

Survey	Parent	Matched	Original observational references
SLACS	57	53	Bolton et al. (2006, 2008); Auger et al. (2009, 2010)
S4TM	38	36	Shu et al. (2015, 2017)
BELLS	21	17	Brownstein et al. (2012)
BELLS GALLERY	14	1	Shu et al. (2016a,b)
Total	130	107	

the maximum is 4.55. A global power-law tracer thus extrapolates the fitted inner profile over the region contributing to the aperture dispersion. Chen et al. (2019) report relative statistical slope errors below 1%, while their population dispersion 0.085 describes variation between galaxies. Individual error estimates and imaging-fit records are unavailable for the supplied slopes. The analytic aperture prediction is

$$\sigma_{\rm th}^2 = \frac{c^2}{2\sqrt{\pi}}\frac{\theta_E}{R_{\rm SN}}F(\gamma,\delta_i,\beta)\left(\frac{\theta_{\rm ap}}{\theta_E}\right)^{2-\gamma}, \tag{3}$$

where c is the speed of light, angular radii in the prefactor are in radians, and F is given in Appendix A.

For comparison, we use a Hernquist tracer (Hernquist 1990), with projected half-light radius R_e corresponding to the catalogue angular radius $\theta_{\rm eff}$,

$$\nu(r) \propto \frac{1}{r(r+a)^3}, \qquad a = R_e/1.81527, \tag{4}$$

and the exact Abel deprojection of the projected Sérsic form

$$I(R) \propto \exp[-b_n(R/R_e)^{1/n}], \qquad \nu(r) = -\frac{1}{\pi}\int_r^\infty \frac{{\rm d}I/{\rm d}R}{\sqrt{R^2-r^2}}\,{\rm d}R. \tag{5}$$

The constant b_n imposes the projected half-light condition. We use $n = 4$ for the main Sérsic comparison and $n = 2$ and 6 as sensitivity scenarios. These models use the measured effective radii and assumed profile families, replacing the individual inner-slope constraint used by the power-law model. The comparison therefore tests the overall photometric representation. A common imaging likelihood would be required to isolate curvature while retaining identical photometric information.

For each curved tracer, we solve the spherical Jeans equation (Binney 1980; Mamon & Łokas 2005),

$$\nu(r)\sigma_r^2(r) = r^{-2\beta}\int_r^\infty \nu(s)\frac{GM(s)}{s^2}s^{2\beta}\,{\rm d}s, \tag{6}$$

normalising the power-law mass by its Einstein projected mass and the stated $R_{\rm SN}$. We then project the pressure and integrate the line-of-sight second moment over the circular aperture. Both the moment numerator and light denominator use the same aperture weighting. The baseline has no PSF convolution; additional cases use an assumed circular Gaussian PSF with $\sigma_{\rm PSF} = 0.64$ arcsec. This assumed seeing value defines a PSF sensitivity scenario for the catalogue.

We evaluate the numerical integrals on 4096 logarithmic radial nodes, with 128 Abel and 192 angular quadrature nodes. A Chebyshev representation accelerates repeated evaluations. Appendix C describes direct-quadrature comparisons, independent off-grid tests, and checks at doubled resolution.

2.5. Conditional likelihood and parameter domain

In matched-catalogue order, the magnitude-difference covariance is

$$\mathbf{C}_{\Delta m} = \mathbf{C}_{ll} + \mathbf{C}_{ss} - \mathbf{C}_{ls} - \mathbf{C}_{ls}^T. \tag{7}$$

First-order derivatives propagate it to $\mathbf{C}_R$ and then to $\mathbf{C}_{\sigma,\rm SN}$ (Appendix B). For the dispersion residual $\mathbf{D} = \boldsymbol{\sigma}_{\rm ap} - \boldsymbol{\sigma}_{\rm th}$,

$$\mathbf{C}_{\rm tot} = \mathbf{C}_{\sigma,\rm SN} + {\rm diag}\left[\sigma_{\rm stat}^2 + (0.03\sigma_{\rm ap})^2 + (f_{\rm int}\sigma_{\rm ap})^2\right], \tag{8}$$

and

$$\ln\mathcal{L} = -\tfrac{1}{2}[\mathbf{D}^T\mathbf{C}_{\rm tot}^{-1}\mathbf{D} + \ln\det\mathbf{C}_{\rm tot} + N\ln(2\pi)]. \tag{9}$$

The propagated covariance is recomputed as the model changes, with all cross-system terms retained and Cholesky factorisation used for evaluation. The 3% term is phenomenological and does not explicitly model coherent calibration or geometry. The fractional-scatter parameter $f_{\rm int}$ is distinct from δ_i. Its observational dispersion anchor is held fixed during parameter inference.

This is a conditional Gaussian approximation, not a latent-supernova likelihood. In particular, the nonlinear transformation of uncertain magnitudes can be non-Gaussian when R is poorly constrained. Section 3.4 quantifies the sensitivity to nonlinear propagation in poorly constrained systems.

The prior boxes are $1 < \gamma_0 < 3$, $-2 < \gamma_z < 2$, $-1 < \beta < 1$, and $0 < f_{\rm int} < 1$, with $1.05 < \gamma_i < 2.95$ at every lens. The Gaussian free-β comparison replaces the uniform beta density by $\mathcal{N}(0.18, 0.13^2)$ truncated to $(-1, 1)$. The historical implementation in Chen et al. (2019) instead restricted this Gaussian to its mean plus or minus two standard deviations, $[-0.08, 0.44]$. Our broad-support Gaussian comparison is consequently not an exact reproduction of that historical prior. For the analytic global power-law moments we additionally impose $1 < \delta_i < 3$, $1 < \xi_i < 3$, and $\xi_i > 2\beta$, where $\xi_i = \gamma_i + \delta_i - 2$. These are convergence conditions, not a proof of a non-negative phase-space distribution function. The curved-tracer integrals are assessed with their own finite-moment domain, without imposing the global power-law ξ restrictions.

We determine each maximum-likelihood estimate (MLE) with three independent differential-evolution searches followed by Nelder–Mead refinement. The maxima must agree within 0.05 in $\ln\mathcal{L}$ and satisfy the nested likelihood ordering. Each main posterior is sampled with four independent `emcee` ensembles (Foreman-Mackey et al. 2013) of 32 walkers, discarding 3000 initial steps. We extend the ensembles according to the convergence criteria in Appendix C; accepted lengths are listed in Table E.1.

For k fitted parameters, the Akaike information criterion (AIC) is ${\rm AIC} = 2k - 2\ln\mathcal{L}_{\rm max}$ and the Bayesian information criterion (BIC) is ${\rm BIC} = k\ln N - 2\ln\mathcal{L}_{\rm max}$. We report

$$\Delta{\rm BIC}_b = {\rm BIC}_{\beta=b} - {\rm BIC}_{\rm free}, \tag{10}$$

so positive values favour the free model under this criterion. BIC with N lens systems is an approximate description for these correlated, model-covariance-dependent data, not a calibrated significance or probability. Flat and Gaussian free-β priors share the likelihood maximum and hence the likelihood-based criteria. We compare criteria only within the same data set.

3. Results and discussion

3.1. Tracer-dependent likelihood comparisons

Table 2 and Fig. 1 compare likelihood maxima on the newly matched 107-lens catalogue. The individual-slope power-law tracer yields $\Delta\mathrm{BIC}_0 = 6.74$ and $\Delta\mathrm{BIC}_{0.18} = 41.51$. The corresponding values are −3.89 and −2.26 for Hernquist, and −4.67 and −4.30 for Sérsic $n = 4$. The criterion preference for freeing the common anisotropy therefore depends on the assumed light profile.

The isotropic Sérsic $n = 4$ likelihood is −513.79, compared with −514.03 for the free power-law case, despite fitting one fewer parameter. The $n = 2$ case retains $\Delta\mathrm{BIC}_{0.18} = 6.61$, while its isotropy comparison is only 0.24. For $n = 6$ the differences are −4.16 and −4.55. The free MLE beta changes from −0.54 at $n = 2$ to −0.01 at $n = 4$ and +0.31 at $n = 6$. A difference near $-\ln 107 = -4.67$ corresponds to nearly equal likelihood maxima, with the parameter-count penalty favouring the fixed case.

3.2. Slope shifts and posterior dependence

The power-law MLE zero-redshift intercept changes from 2.194 with free beta to 2.019 and 1.962 at fixed 0 and 0.18. For Hernquist the values are 2.186, 2.125, and 2.076; for Sérsic $n = 4$ they are 2.128, 2.126, and 2.084. At the pivot redshift the power-law values are 2.018, 1.810, and 1.740, compared with 1.995, 1.993, and 1.957 for Sérsic $n = 4$. Thus the sensitivity is also present at a redshift represented in the sample.

Table 3 and Fig. 2 give posterior medians and central 68% intervals under both free-beta priors and both fixed hypotheses. Every pivot interval is computed from the joint draws. Figure 3 shows the common parameters across the four power-law treatments. Appendix F shows the complete joint posterior distributions. The free-flat power-law posterior has 1.48% of beta draws within 0.02 of its lower prior bound −1.

3.3. Controlled recovery and tracer mismatch

We generate correlated dispersion vectors for four MLE-based generating models: isotropic power law, free-beta power law, isotropic Sérsic $n = 4$, and isotropic Hernquist. There are 200 predetermined realisations per truth. Each vector is fitted with free beta, fixed 0, and fixed 0.18, under both its true tracer and an alternative family. All 800 realisations and their 4800 fitted models pass the specified optimisation checks. Null and alternative experiments use the same noise prescription. The generating dispersion sets the fractional-scatter variance anchor, which remains fixed during fitting. The conditional noise model is held fixed across these paired comparisons.

Table 4 and Fig. 4 report paired comparisons. Under isotropic Sérsic $n = 4$ and Hernquist truths, fitting the power-law tracer produces $\Delta\mathrm{BIC}_0 > 6$ in 60.0% and 50.0% of realisations, respectively, with mean effective-beta shifts of −0.547 and −0.492. Matched-tracer isotropic fits cross this threshold in only 0.5% and 1.0% of those realisations. Under an isotropic power-law truth, the matched power-law rate is 0.5%. The free-beta power-law truth gives 67.5% recovery above the threshold under its own family. These frequencies depend on the chosen MLE truths, noise model, catalogue, and threshold. The cross-tracer fits quantify how the specified mismatch can generate an effective anisotropy preference.

3.4. Distance uncertainty and matching

The linearised fractional distance-ratio uncertainty has median 0.0651, reaches 0.443, and exceeds 0.20 in 12 systems. In 50000 correlated Gaussian magnitude-difference draws, the largest fraction yielding $R \leq 0.001$ is 3.6% (SDSSJ2122+0409). These diagnostic draws quantify nonlinear propagation about the observed magnitudes; their interpretation is discussed in Sect. 3.6. Excluding systems above 20% linearised fractional distance-ratio uncertainty leaves 95 lenses. The power-law isotropy comparison becomes 2.51; Hernquist and Sérsic $n = 4$ give −4.04 and −4.54. Exclusion changes both uncertainty and sample design, so the difference cannot be attributed uniquely to nonlinear propagation. The likelihood approximation and its implications are discussed in Sect. 3.6.

Appendix D gives corrections for finite redshift mismatch under smooth flat backgrounds with $\Omega_m = 0.25$, 0.30, and 0.35, together with a heliocentric-redshift transfer check. These corrections transform both the ratios and covariance. The dependence on the assumed background is included in these comparisons.

We also repeat the two-stage matching at tolerances 0.03 and 0.07, obtaining 104 and 111 lenses. Their power-law isotropy comparisons are 3.69 and 6.90. Reversing the SN input ordering gives another 107-lens catalogue with 78 complete triplets shared with the primary assignment and differences of 6.51 and 41.67 for power law. All three alternatives retain negative isotropy differences for Hernquist and Sérsic $n = 4$ (Table E.2). The principal tracer dependence persists across these assignments, while the strength of the power-law preference varies.

3.5. Photometric uncertainty and structural control

Coherent offsets of each measured δ_i by ±0.05 and ±0.10 are refitted on the same primary catalogue (Table D.1). These assumed offsets probe sensitivity to photometric calibration. We separately integrate over latent slopes with a truncated-normal prior on $(1.001, 2.999)$ and enforce the dynamical convergence domain jointly with the global parameters. Four independent chains use blocked elliptical-slice updates for Gaussian latent coordinates mapped through the truncated-normal cumulative distribution, and global random-walk updates adapted only during burn-in (Murray et al. 2010).

The smallest-error scenario uses $\sigma_{\delta,i} = 0.01\widehat{\delta_i}$, motivated by the upper bound on relative statistical errors stated by Chen et al. (2019). We treat this width as a statistical upper-bound scenario. Additional absolute widths 0.05 and 0.10, and width 0.05 with common offset $b \sim \mathcal{N}(0, 0.05^2)$, probe broader modelling uncertainties. The published population scatter 0.085 is not substituted for a measurement error. Table 5 gives the new results. All eight posterior calculations satisfy the diagnostics for the global parameters and every varying latent coordinate. These slope-error priors apply to the power-law tracer; we report the marginal posteriors separately from the likelihood-based BIC comparisons. The effective-beta medians and central 68% intervals under the free-flat prior are $\sigma_\delta = 0.05$: $-0.533^{+0.231}_{-0.275}$;

Table 2. P2 likelihood maxima. Differences are fixed minus free within each tracer scenario on the same 107 lenses. Pivot slopes use $z_p = 0.1959$. PSF and Sérsic-index choices are assumed scenarios.

Tracer	β treatment	$\ln \mathcal{L}_{\max}$	ΔBIC	$\widehat{\gamma_p}$	$\widehat{\beta}$	$\widehat{f_{\rm int}}$
Power law	Free, flat	-514.03	0.00	2.018	-0.569	0.080
Power law	Fixed 0	-519.74	6.74	1.810	0.000	0.086
Power law	Fixed 0.18	-537.12	41.51	1.740	0.180	0.107
Hernquist	Free, flat	-514.37	0.00	2.038	-0.331	0.081
Hernquist	Fixed 0	-514.77	-3.89	1.987	0.000	0.082
Hernquist	Fixed 0.18	-515.58	-2.26	1.948	0.180	0.083
Sérsic $n = 4$	Free, flat	-513.79	0.00	1.995	-0.012	0.080
Sérsic $n = 4$	Fixed 0	-513.79	-4.67	1.993	0.000	0.080
Sérsic $n = 4$	Fixed 0.18	-513.98	-4.30	1.957	0.180	0.081
Sérsic $n = 2$	Free, flat	-514.00	0.00	2.019	-0.541	0.080
Sérsic $n = 2$	Fixed 0	-516.45	0.24	1.881	0.000	0.084
Sérsic $n = 2$	Fixed 0.18	-519.64	6.61	1.808	0.180	0.089
Sérsic $n = 6$	Free, flat	-513.61	0.00	1.977	0.309	0.080
Sérsic $n = 6$	Fixed 0	-513.86	-4.16	2.021	0.000	0.080
Sérsic $n = 6$	Fixed 0.18	-513.67	-4.55	1.999	0.180	0.080
Hernquist + PSF	Free, flat	-513.99	0.00	2.029	-0.282	0.080
Hernquist + PSF	Fixed 0	-514.30	-4.05	1.984	0.000	0.081
Hernquist + PSF	Fixed 0.18	-515.07	-2.51	1.943	0.180	0.082
Sérsic $n = 4$ + PSF	Free, flat	-513.57	0.00	1.991	-0.003	0.080
Sérsic $n = 4$ + PSF	Fixed 0	-513.57	-4.67	1.991	0.000	0.080
Sérsic $n = 4$ + PSF	Fixed 0.18	-513.75	-4.33	1.955	0.180	0.080

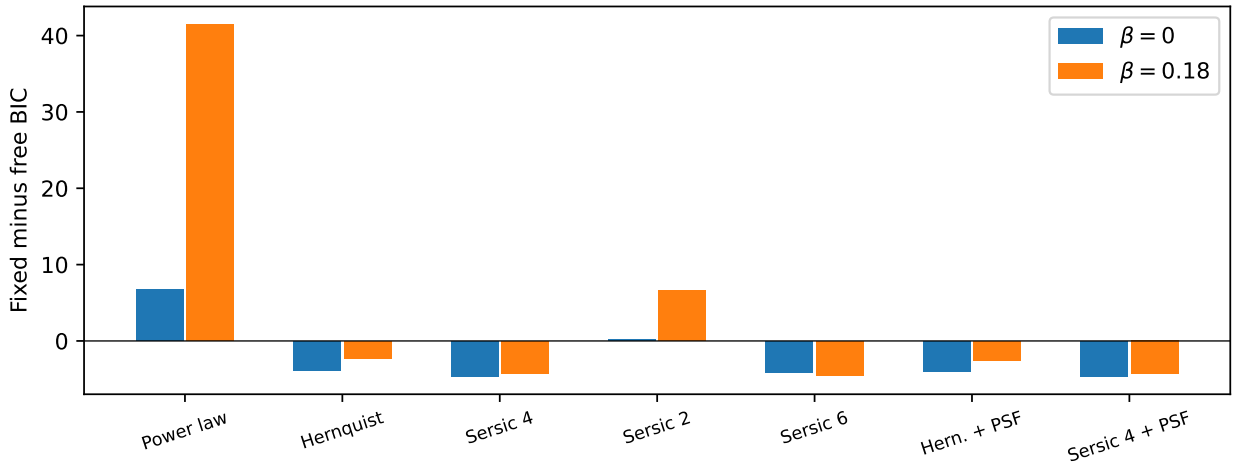


Fig. 1. Tracer and PSF dependence of the fixed-minus-free BIC differences.

$\sigma_\delta = 0.10$: $-0.499^{+0.219}_{-0.278}$; $\sigma_\delta = 0.05$, $\sigma_b = 0.05$: $-0.540^{+0.226}_{-0.267}$; $\sigma_{\delta,i} = 0.01\widehat{\delta_i}$: $-0.541^{+0.232}_{-0.272}$. The four rows quantify sensitivity to the stated slope-error models.

Removing the BELLS family leaves 89 lenses and gives $\Delta\mathrm{BIC}_0 = 7.40$ for power law; removing S4TM leaves 71 lenses and gives -1.82. The main curved-tracer comparisons remain negative. Survey, redshift, and aperture coverage change together, preventing causal attribution to a single survey.

As a three-coefficient structural control, P3 adds $\gamma_s \log_{10} \widetilde{\Sigma}_L$ to Eq. (2), where

$$\widetilde{\Sigma}_L = 166 \, \frac{D_s/(1\,\mathrm{Mpc})}{[D_l/(1\,\mathrm{Mpc})][D_{ls}/(1\,\mathrm{Mpc})]} \left(\frac{\theta_E}{\theta_{\rm eff}}\right)^2 . \tag{11}$$

The predictor distances assume flat $\Omega_m = 0.30$ and $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. This differs from the kinematic predictor used by Chen et al. (2019). The power-law P3 comparisons give 6.87 and 35.19. Its pivot slope additionally conditions on the sample median predictor, and its posterior parameters appear in Table E.3.

3.6. Scope and limitations

All models assume spherical symmetry, a power-law total mass density, and one radially constant beta shared by the lenses. They do not fit flattening, rotation, anisotropy gradients, or population scatter. The curved tracers replace the individual inner-slope information with effective radii and assumed profile families. Original imaging and consistent PSF modelling are needed to compare profiles under a common photometric likelihood. A broadband light profile can also differ from the spectral weighting of the kinematic measurement (Forés-Toribio et al. 2026).

The SN ratios are correlated and use finite redshift matching and first-order covariance propagation; Sect. 3.4 reports the corresponding sensitivity tests. Gaussian magnitude-difference draws describe nonlinear propagation around the observed magnitudes, while a latent-distance analysis would require a speci-

Table 3. P2 posterior medians and central 68% intervals. Fixed beta entries have no posterior interval.

Tracer	β treatment	γ_p	γ_z	β	$f_{\rm int}$	γ_0
Power law	Free, flat	$2.011^{+0.047}_{-0.064}$	$-0.901^{+0.225}_{-0.234}$	$-0.545^{+0.234}_{-0.271}$	$0.082^{+0.012}_{-0.011}$	$2.186^{+0.055}_{-0.068}$
Power law	Free, Gaussian	$1.877^{+0.049}_{-0.055}$	$-0.996^{+0.276}_{-0.251}$	$-0.120^{+0.067}_{-0.076}$	$0.085^{+0.012}_{-0.011}$	$2.071^{+0.071}_{-0.081}$
Power law	Fixed 0	$1.808^{+0.044}_{-0.048}$	$-0.987^{+0.298}_{-0.253}$	0	$0.088^{+0.012}_{-0.011}$	$1.999^{+0.074}_{-0.083}$
Power law	Fixed 0.18	$1.740^{+0.054}_{-0.057}$	$-0.994^{+0.337}_{-0.257}$	0.18	$0.110^{+0.014}_{-0.013}$	$1.930^{+0.088}_{-0.096}$
Hernquist	Free, flat	$2.040^{+0.044}_{-0.057}$	$-0.749^{+0.196}_{-0.193}$	$-0.360^{+0.345}_{-0.381}$	$0.084^{+0.012}_{-0.011}$	$2.186^{+0.065}_{-0.077}$
Hernquist	Free, Gaussian	$1.965^{+0.037}_{-0.040}$	$-0.670^{+0.204}_{-0.202}$	$0.104^{+0.109}_{-0.113}$	$0.085^{+0.012}_{-0.011}$	$2.096^{+0.061}_{-0.064}$
Hernquist	Fixed 0	$1.987^{+0.028}_{-0.029}$	$-0.698^{+0.198}_{-0.196}$	0	$0.084^{+0.012}_{-0.011}$	$2.123^{+0.052}_{-0.053}$
Hernquist	Fixed 0.18	$1.947^{+0.030}_{-0.030}$	$-0.652^{+0.205}_{-0.203}$	0.18	$0.085^{+0.012}_{-0.011}$	$2.075^{+0.055}_{-0.056}$
Sérsic $n = 4$	Free, flat	$2.009^{+0.053}_{-0.066}$	$-0.682^{+0.188}_{-0.184}$	$-0.109^{+0.322}_{-0.454}$	$0.083^{+0.012}_{-0.011}$	$2.143^{+0.068}_{-0.083}$
Sérsic $n = 4$	Free, Gaussian	$1.961^{+0.035}_{-0.038}$	$-0.652^{+0.193}_{-0.190}$	$0.155^{+0.113}_{-0.118}$	$0.083^{+0.012}_{-0.011}$	$2.089^{+0.055}_{-0.059}$
Sérsic $n = 4$	Fixed 0	$1.992^{+0.024}_{-0.025}$	$-0.676^{+0.184}_{-0.186}$	0	$0.082^{+0.012}_{-0.011}$	$2.124^{+0.046}_{-0.047}$
Sérsic $n = 4$	Fixed 0.18	$1.955^{+0.026}_{-0.026}$	$-0.648^{+0.192}_{-0.193}$	0.18	$0.083^{+0.012}_{-0.011}$	$2.082^{+0.049}_{-0.050}$

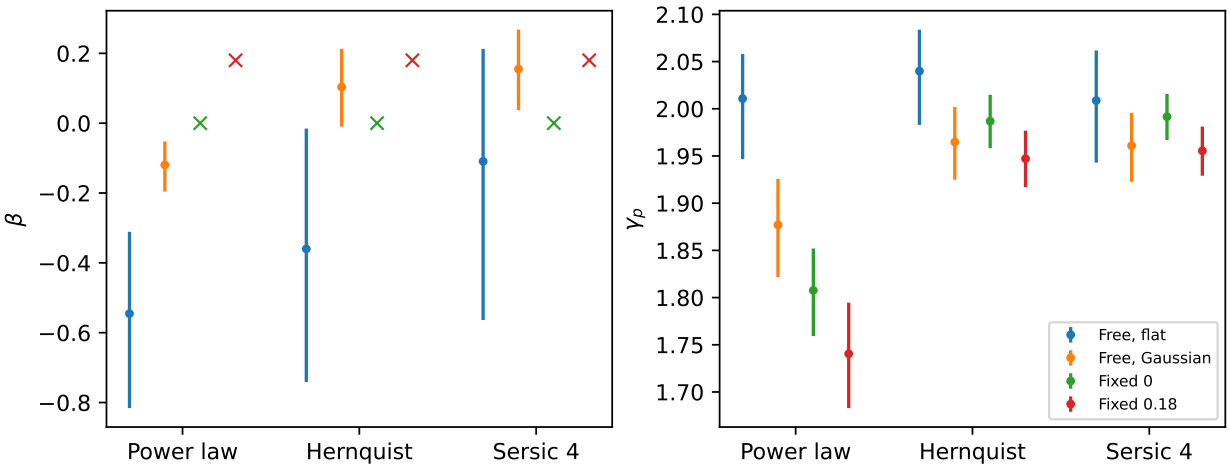


Fig. 2. Effective beta and pivot-slope posteriors versus tracer and anisotropy treatment. Crosses denote fixed beta values.

Table 4. Conditional recovery experiments, 200 realisations per truth. Threshold frequencies include Wilson 95% intervals in percent. Biases refer to each stated generative truth.

Generating truth	Fitted tracer	$\langle \Delta\mathrm{BIC}_0 \rangle$	$P(\Delta\mathrm{BIC}_0 > 6)$ (%)	$\langle \widehat{\beta} - \beta_{\rm true} \rangle$	$\langle \widehat{\gamma}_p - \gamma_{p,\rm true} \rangle$
Hernquist, isotropic	Hernquist	-3.63	1.0 [0.3, 3.6]	-0.047	-0.008
Hernquist, isotropic	Power law	6.88	50.0 [43.1, 56.9]	-0.492	-0.008
Power law, free MLE	Power law	9.81	67.5 [60.7, 73.6]	-0.029	-0.007
Power law, free MLE	Sérsic $n = 4$	-2.69	0.0 [0.0, 1.9]	0.026	0.036
Power law, isotropic	Power law	-3.56	0.5 [0.1, 2.8]	-0.005	-0.010
Power law, isotropic	Sérsic $n = 4$	-1.99	3.0 [1.4, 6.4]	0.321	0.125
Sérsic $n = 4$, isotropic	Power law	8.44	60.0 [53.1, 66.5]	-0.547	0.000
Sérsic $n = 4$, isotropic	Sérsic $n = 4$	-3.56	0.5 [0.1, 2.8]	-0.085	-0.006

fied prior and joint treatment of the supernova data. The additional 3% variance term and fitted scatter give a phenomenological uncertainty model. The fitted γ_z characterises the selected, heterogeneous sample (Bolton et al. 2012; Sonnenfeld et al. 2013; Chen et al. 2019), and γ_0 is its extrapolated zero-redshift intercept. The inference targets dynamical model sensitivities; cosmological parameters, including H_0, are not fitted.

The assumed Sérsic indices, PSF values, and broader latent-slope widths define sensitivity scenarios in the absence of individual image-fitting and spectroscopic-extraction likelihoods. Within this scope, the likelihood, recovery, and sensitivity comparisons quantify how light-profile assumptions can affect a common-anisotropy preference. Interpreting that preference as an individual orbital property, or attributing the observed residuals to a unique physical cause, requires additional photometric and kinematic constraints.

4. Conclusions

We quantified the dependence of anisotropy and density-slope inference on tracer-profile assumptions using 107 galaxy-scale lenses and distinct Pantheon+ counterparts. The main findings are:

1. The baseline individual-slope power-law tracer yields $\Delta\mathrm{BIC}_0 = 6.74$ and $\Delta\mathrm{BIC}_{0.18} = 41.51$ when each specified fixed reference is compared with a common free anisotropy parameter.

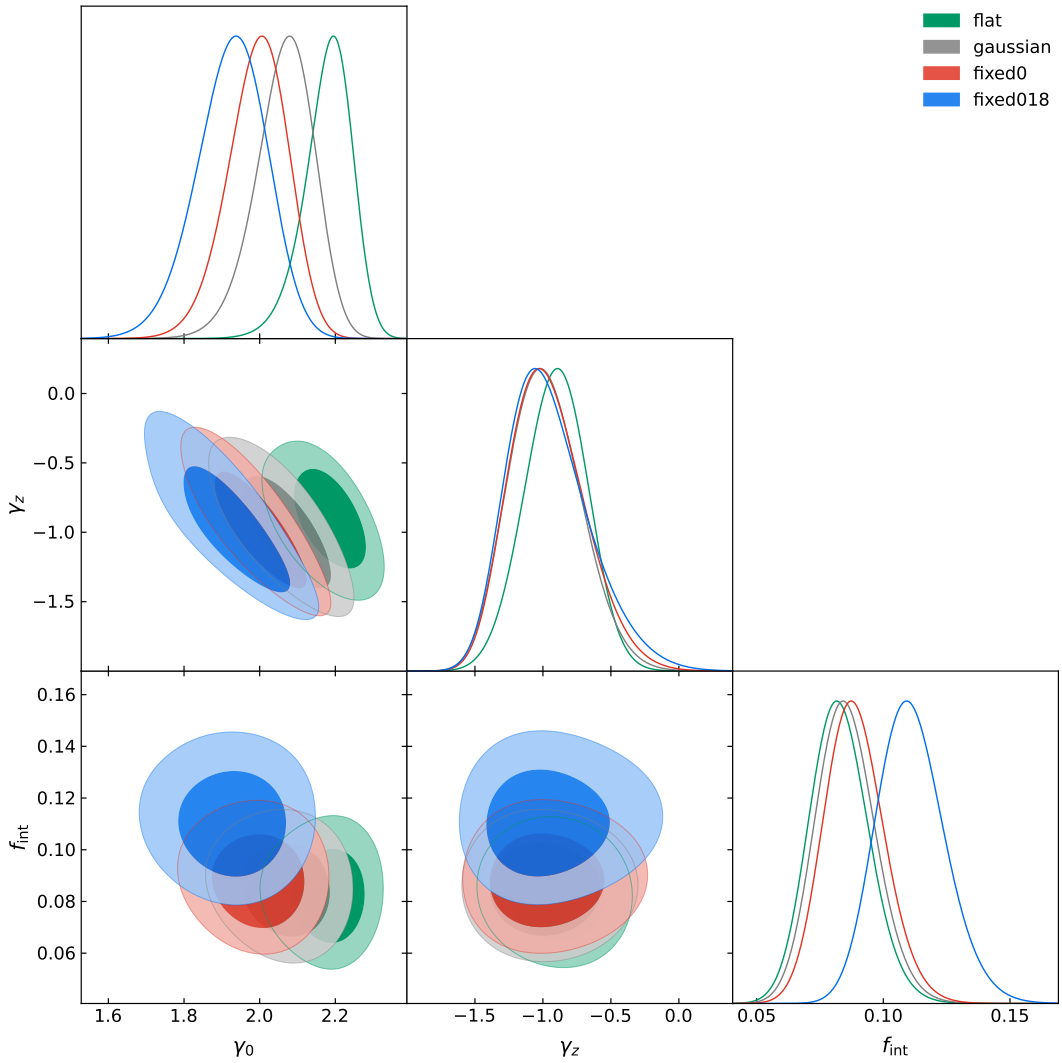


Fig. 3. Power-law P2 posterior comparison under all four anisotropy treatments. Only common sampled parameters are displayed; free beta is marginalised over.

Table 5. Power-law posterior sensitivity to slope-error scenarios. Entries are medians and central 68% intervals. The relative-1% row is a statistical upper-bound scenario; the other widths are assumed sensitivity inputs.

Slope-error scenario	β treatment	γ_p	γ_z	β	$f_{\rm int}$
$\sigma_\delta = 0.05$	Free, flat	$2.008^{+0.048}_{-0.065}$	$-0.904^{+0.225}_{-0.235}$	$-0.533^{+0.231}_{-0.275}$	$0.082^{+0.012}_{-0.011}$
$\sigma_\delta = 0.05$	Fixed 0	$1.810^{+0.043}_{-0.047}$	$-1.000^{+0.296}_{-0.254}$	0	$0.085^{+0.012}_{-0.011}$
$\sigma_\delta = 0.10$	Free, flat	$2.001^{+0.050}_{-0.064}$	$-0.909^{+0.226}_{-0.241}$	$-0.499^{+0.219}_{-0.278}$	$0.080^{+0.012}_{-0.011}$
$\sigma_\delta = 0.10$	Fixed 0	$1.816^{+0.041}_{-0.045}$	$-1.022^{+0.294}_{-0.255}$	0	$0.076^{+0.013}_{-0.013}$
$\sigma_\delta = 0.05$, $\sigma_b = 0.05$	Free, flat	$2.007^{+0.048}_{-0.065}$	$-0.914^{+0.237}_{-0.256}$	$-0.540^{+0.226}_{-0.267}$	$0.082^{+0.012}_{-0.011}$
$\sigma_\delta = 0.05$, $\sigma_b = 0.05$	Fixed 0	$1.866^{+0.044}_{-0.051}$	$-0.954^{+0.270}_{-0.251}$	0	$0.084^{+0.012}_{-0.011}$
$\sigma_{\delta,i} = 0.01\widehat{\delta_i}$	Free, flat	$2.010^{+0.048}_{-0.064}$	$-0.904^{+0.225}_{-0.232}$	$-0.541^{+0.232}_{-0.272}$	$0.082^{+0.012}_{-0.011}$
$\sigma_{\delta,i} = 0.01\widehat{\delta_i}$	Fixed 0	$1.808^{+0.044}_{-0.048}$	$-0.990^{+0.297}_{-0.252}$	0	$0.088^{+0.012}_{-0.011}$

2. Hernquist and Sérsic $n = 4$ alternatives remove this criterion advantage and reduce the slope changes induced by fixing beta. The contrasting results quantify sensitivity to the photometric representation, which uses different information in the power-law and curved-tracer cases.
3. The comparison also depends on distance uncertainty, survey composition, and the input slopes. The 95-lens lower-uncertainty subset gives a power-law isotropy difference of 2.51. The principal tracer dependence persists across the finite-matching and alternative-assignment checks, while the strength of the power-law preference varies.
4. Under the specified conditional noise model, fitting a power-law tracer to isotropic Sérsic $n = 4$ and Hernquist simulations produces $\Delta\mathrm{BIC}_0 > 6$ in 60.0% and 50.0% of realisations, respectively. The corresponding matched-tracer frequencies are 0.5% and 1.0%. These controlled experiments show how tracer mismatch can generate an effective anisotropy preference.

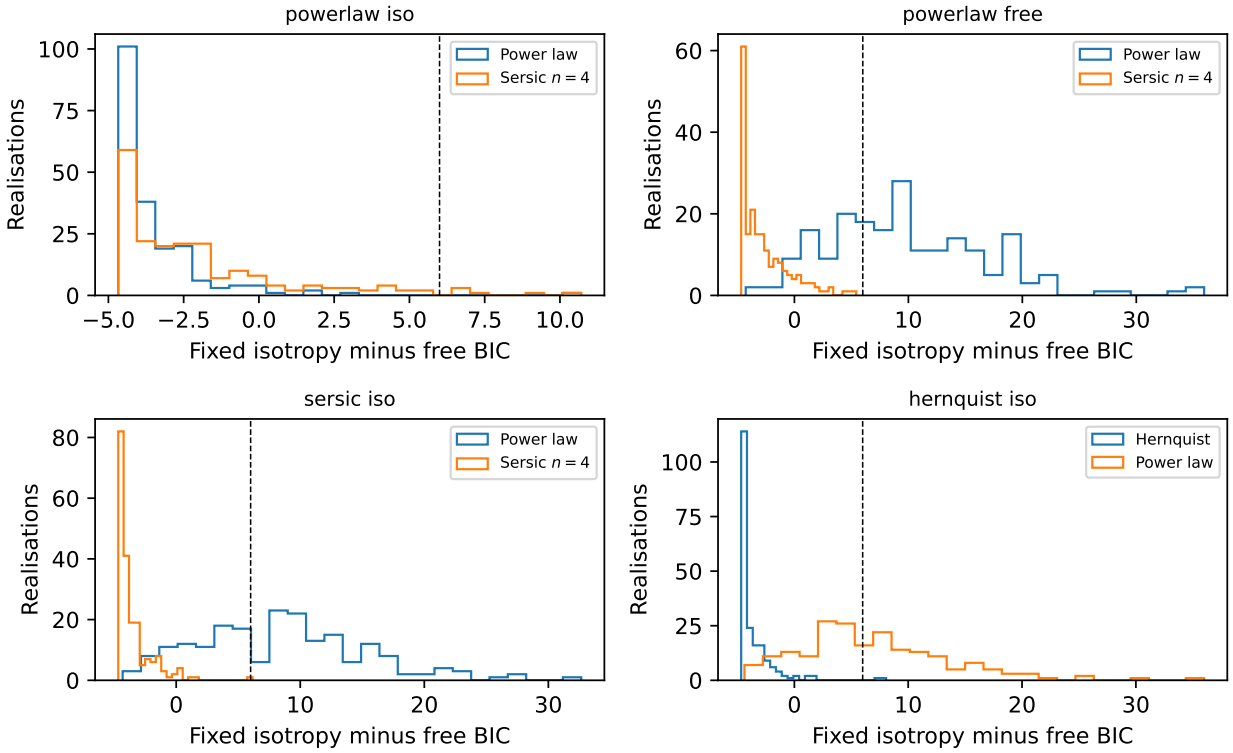


Fig. 4. Truth-matched and alternative-tracer recovery distributions for the same simulated vectors. The line at 6 is a descriptive BIC threshold, not a universal significance.

The combined comparisons motivate joint assessment of anisotropy, light-profile representation, photometric uncertainty, and aperture weighting. They provide a quantitative basis for testing light-profile assumptions before interpreting a fitted common beta as a population orbital property. Consistent imaging constraints and spatially resolved spectroscopy can further separate tracer structure from orbital anisotropy.

Data availability

The Pantheon+ data and covariance are available from the official release at https://github.com/PantheonPlusSH0ES/DataRelease. Original lens surveys are cited in Table 1, and the compiled lens parameters are listed in Table A1 of Chen et al. (2019). The individual luminosity-slope measurements were provided by Y. Chen. The repository for our analysis code and data products is https://github.com/HUJIAN0000/sgl-sn-anisotropy. The revised code, matched catalogue, and derived summary tables accompanying this manuscript will be provided in an updated release at the same address.

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Appendix A: Analytic power-law normalisation

With $\xi = \gamma + \delta_i - 2$, the dimensionless function in Eq. (3) is

$$F = \frac{3-\delta_i}{(\xi-2\beta)(3-\xi)}\left[\frac{\Gamma((\xi-1)/2)}{\Gamma(\xi/2)} - \beta\frac{\Gamma((\xi+1)/2)}{\Gamma((\xi+2)/2)}\right] \times \frac{\Gamma(\gamma/2)\Gamma(\delta_i/2)}{\Gamma((\gamma-1)/2)\Gamma((\delta_i-1)/2)}. \quad (A.1)$$

For $\gamma = \delta_i = 2$ and $\beta = 0$, $F = 1/(2\sqrt{\pi})$, giving $\sigma^2 = c^2\theta_E/(4\pi R)$, the singular-isothermal-sphere normalisation. The numerical power-law engine reproduces this limit and general convergent analytic moments. Formal analytic continuations outside the convergent domain are not used for the main posterior.

Appendix B: Covariance propagation

Let $q_i = 1 - R_i$. The derivatives are

$$\frac{\partial R_i}{\partial \Delta m_i} = -\frac{\ln 10}{5} q_i, \qquad \frac{\partial \sigma_i}{\partial R_i} = -\frac{\sigma_i}{2R_i}. \quad (B.1)$$

The diagonal Jacobians give $\mathbf{C}_R = \mathbf{J}_{Rm}\mathbf{C}_{\Delta m}\mathbf{J}_{Rm}^T$ and $\mathbf{C}_{\sigma,\mathrm{SN}} = \mathbf{J}_{\sigma R}\mathbf{C}_R\mathbf{J}_{\sigma R}^T$. The latter is recalculated at every likelihood evaluation. This first-order construction retains covariance but does not represent finite-redshift matching errors or a nonlinear distance posterior.

For a fiducial-assisted redshift correction, define $q_i^{\mathrm{match,fid}}$ by applying the matching formula to noiseless model magnitudes at the actual counterpart redshifts and $q_i^{\mathrm{target,fid}} = \chi_{l,i}/\chi_{s,i}$. With $a_i = q_i^{\mathrm{target,fid}}/q_i^{\mathrm{match,fid}}$, the check uses $R_i' = 1 - a_i(1-R_i)$ and $(C_R')_{ij} = a_i(C_R)_{ij}a_j$. The assumed background dependence is explicit.

Appendix C: Numerical verification and sampling diagnostics

The radial calculations are checked against doubled radial resolution, a factor-ten extension of the radial limits, and doubled Abel/angular nodes. Each curved-tracer case is evaluated at its three likelihood maxima and 50 independent feasible perturbations near the free maximum. We require $|\Delta \ln \mathcal{L}| < 0.1$ and maximum relative dispersion changes below 5×10^{-4}. Differences smaller than the numerical uncertainty are interpreted as effectively equal likelihoods.

To accelerate sampling, a tensor Chebyshev representation of $\log(\sigma^2 R/A(\gamma))$ is built from direct integrations, where $A(\gamma) = 2\Gamma(\gamma/2)/[\sqrt{\pi}\Gamma((\gamma-1)/2)]$ is evaluated analytically. Removing this known factor improves behaviour near the lower gamma boundary. The representation is checked using 120 independent off-grid parameter sets across the supported gamma and beta range, requiring relative dispersion error below 2×10^{-6}, plus explicit likelihood checks against the direct engine. The cache includes the input, profile, grid and implementation hashes.

Posterior checks include rank-normalised and folded split $\widehat{R} < 1.01$ (Vehtari et al. 2021), autocorrelation-based effective sample size (ESS) at least 2000, retained chain lengths exceeding 100τ, and changes in estimated τ below 5%. Separate ensembles supply independent runs; interacting walkers are not declared independent chains. Chain lengths, diagnostic arrays and trace plots are retained in the reproduction material. The same convergence thresholds are applied throughout the sampling. All 16 main and eight latent-slope posterior calculations pass the stated diagnostics. The largest reported rank-normalised split $\widehat{R}$ is 1.0080, and the smallest approximate ESS is 25868. The curved-tracer numerical audit additionally evaluates 24 retained posterior draws per tracer/prior combination, 192 points in total, against doubled quadrature resolution. These checks supplement the likelihood-maximum and off-grid interpolation tests.

Appendix D: Sensitivity comparisons

Table D.1 compares each beta hypothesis within its own stated sample and scenario. Absolute BIC values are not compared between different data memberships. Coherent delta offsets are assumed perturbations; mismatch corrections use stated fiducial backgrounds. The separate P3 control uses the power-law tracer and one additional structural coefficient.

Appendix E: Sampling convergence diagnostics and matching variants

For each main posterior we use four independently initialised ensembles; each latent-slope posterior uses four independent chains. Main ensembles contain 15000 or 25000 steps with 3000 discarded, and latent-slope chains contain 80000–320000 steps with 5000 discarded. Depending on the beta treatment and common-offset scenario, the latent analyses have 110–112 varying coordinates. Table E.1 lists the accepted lengths, maximum rank-normalised split $\widehat{R}$, and minimum autocorrelation-based approximate ESS. The latent diagnostics include all varying slope coordinates and the common offset where fitted.

We retain the matched supernova row indices, covariance blocks, configurations, chain checkpoints, and diagnostic arrays to support reproducibility. The magnitude covariance is symmetrised as $(C + C^T)/2$ to remove the published rounding-level asymmetry (maximum 3×10^{-8}). Table E.2 gives the alternative assignments, and Table E.3 reports the P3 posterior parameters. Approximate ESS is used alongside chain-length, autocorrelation-stability, between-run, and trace checks.

Appendix F: Joint posterior distributions

The following triangles show all fitted global parameters for each of the four anisotropy treatments. Every distribution is computed from the retained samples for the primary 107-lens catalogue. The black dashed line on each diagonal marks the full-sample median; the annotations give the median and central 68% interval. The sienna and tan bands show the 16th–84th and 2.275th–97.725th percentile intervals, respectively, labelled 1σ and 2σ for display. The inner and outer blue contours enclose 68% and 95% of each two-dimensional marginal density; these joint probabilities are distinct from the diagonal equal-tailed intervals. GetDist provides the smoothed densities with relative smoothing scales 0.5 in one dimension and 0.7 in two dimensions. All retained draws are used without random subsampling. The density estimator is supplied with the actual hard prior ranges to avoid smoothing into excluded parameter values. Smoothing is a display operation and does not enter the reported numerical intervals.

Figures F.1 and F.2 show the P2 power-law cases; Figs. F.3 and F.4 show Hernquist; Figs. F.5 and F.6 show Sérsic $n = 4$. Figures F.7, F.8, and F.9 give the P3 controls. Fixed-beta triangles omit beta because it is not sampled. The plotted γ_0 remains the zero-redshift intercept, and f_{int} is fractional dispersion scatter. The panels display the parameter correlations for each tracer and

Table D.1. Sensitivity comparisons. Positive fixed-minus-free differences favour the free model within the given row. Subset differences conflate information loss, selection and changed design; they are not estimates of a causal survey effect.

Scenario	Tracer	N	$\Delta\mathrm{BIC}_0$	$\Delta\mathrm{BIC}_{0.18}$
$\sigma_R/R \leq 0.20$	Power law	95	2.51	29.90
$\sigma_R/R \leq 0.20$	Hernquist	95	-4.04	-2.86
$\sigma_R/R \leq 0.20$	Sérsic $n = 4$	95	-4.54	-4.40
Exclude BELLS family	Power law	89	7.40	38.86
Exclude BELLS family	Hernquist	89	-2.62	-0.60
Exclude BELLS family	Sérsic $n = 4$	89	-4.09	-3.21
Exclude S4TM	Power law	71	-1.82	9.69
Exclude S4TM	Hernquist	71	-4.10	-3.54
Exclude S4TM	Sérsic $n = 4$	71	-4.17	-4.26
Fiducial mismatch ($\Omega_m = 0.25$)	Power law	107	6.68	41.22
Fiducial mismatch ($\Omega_m = 0.25$)	Hernquist	107	-3.84	-2.17
Fiducial mismatch ($\Omega_m = 0.25$)	Sérsic $n = 4$	107	-4.67	-4.26
Fiducial mismatch ($\Omega_m = 0.30$)	Power law	107	6.68	41.23
Fiducial mismatch ($\Omega_m = 0.30$)	Hernquist	107	-3.84	-2.17
Fiducial mismatch ($\Omega_m = 0.30$)	Sérsic $n = 4$	107	-4.67	-4.26
Fiducial mismatch ($\Omega_m = 0.35$)	Power law	107	6.68	41.23
Fiducial mismatch ($\Omega_m = 0.35$)	Hernquist	107	-3.84	-2.17
Fiducial mismatch ($\Omega_m = 0.35$)	Sérsic $n = 4$	107	-4.67	-4.26
Heliocentric transfer ($\Omega_m = 0.30$)	Power law	107	6.64	41.17
Heliocentric transfer ($\Omega_m = 0.30$)	Hernquist	107	-3.85	-2.19
Heliocentric transfer ($\Omega_m = 0.30$)	Sérsic $n = 4$	107	-4.67	-4.27
Slope offset $\Delta\delta = -0.10$	Power law	107	33.10	97.98
Slope offset $\Delta\delta = -0.05$	Power law	107	16.09	69.47
Slope offset $\Delta\delta = +0.05$	Power law	107	2.21	21.41
Slope offset $\Delta\delta = +0.10$	Power law	107	0.06	10.58

anisotropy treatment. The P3 posterior uses the lensing-based predictor defined in the structural-control subsection, a uniform $-1.5 < \gamma_s < 1.5$ prior, and the same sampling and convergence requirements as P2. It remains a power-law structural control and does not extend the curved-tracer analysis to P3.

Table E.1. Accepted posterior diagnostics. Each row contains four independent ensembles or chains. The minimum approximate ESS and maximum rank-normalised split $\widehat{R}$ include every varying latent coordinate for the slope-error products.

Analysis	Model or slope-error scenario	β treatment	Steps	max $\widehat{R}$	Min. approx. ESS
Main	Power law	Free, flat	25000	1.0057	44560
Main	Power law	Free, Gaussian	25000	1.0055	56369
Main	Power law	Fixed 0	25000	1.0045	68024
Main	Power law	Fixed 0.18	25000	1.0032	65834
Main	Hernquist	Free, flat	25000	1.0079	47316
Main	Hernquist	Free, Gaussian	25000	1.0041	57722
Main	Hernquist	Fixed 0	25000	1.0026	71119
Main	Hernquist	Fixed 0.18	15000	1.0080	38886
Main	Sérsic $n = 4$	Free, flat	25000	1.0064	41319
Main	Sérsic $n = 4$	Free, Gaussian	25000	1.0041	56329
Main	Sérsic $n = 4$	Fixed 0	15000	1.0079	38693
Main	Sérsic $n = 4$	Fixed 0.18	15000	1.0045	39428
Main	P3 power law	Free, flat	25000	1.0069	34611
Main	P3 power law	Free, Gaussian	25000	1.0047	45247
Main	P3 power law	Fixed 0	25000	1.0037	53122
Main	P3 power law	Fixed 0.18	25000	1.0044	50849
Latent slopes	$\sigma_{\delta,i} = 0.01\widehat{\delta_i}$	Free, flat	80000	1.0001	50518
Latent slopes	$\sigma_{\delta,i} = 0.01\widehat{\delta_i}$	Fixed 0	80000	1.0001	72233
Latent slopes	$\sigma_\delta = 0.05$	Free, flat	80000	1.0001	43017
Latent slopes	$\sigma_\delta = 0.05$	Fixed 0	120000	1.0001	73788
Latent slopes	$\sigma_\delta = 0.10$	Free, flat	120000	1.0001	39199
Latent slopes	$\sigma_\delta = 0.10$	Fixed 0	320000	1.0001	67473
Latent slopes	$\sigma_\delta = 0.05$, $\sigma_b = 0.05$	Free, flat	80000	1.0001	42228
Latent slopes	$\sigma_\delta = 0.05$, $\sigma_b = 0.05$	Fixed 0	80000	1.0002	25868

Table E.2. Alternative matching assignments. Comparisons are within each row; shared triplets use original lens and SN identifiers.

Assignment	Tracer	N	Shared triplets	$\Delta\mathrm{BIC}_0$	$\Delta\mathrm{BIC}_{0.18}$
Tolerance 0.03	Power law	104	92	3.69	35.32
Tolerance 0.03	Hernquist	104	92	-4.32	-3.12
Tolerance 0.03	Sérsic $n = 4$	104	92	-4.56	-4.56
Tolerance 0.07	Power law	111	81	6.90	41.99
Tolerance 0.07	Hernquist	111	81	-4.32	-3.04
Tolerance 0.07	Sérsic $n = 4$	111	81	-4.68	-4.56
Reversed SN order	Power law	107	78	6.51	41.67
Reversed SN order	Hernquist	107	78	-3.75	-2.00
Reversed SN order	Sérsic $n = 4$	107	78	-4.66	-4.22

Table E.3. P3 power-law posterior medians and central 68% intervals, at the sample median redshift and median structural predictor.

β treatment	γ_p	γ_0	γ_z	γ_s	f_{int}
Free, flat	$2.014^{+0.050}_{-0.069}$	$2.142^{+0.101}_{-0.112}$	$-0.841^{+0.262}_{-0.276}$	$-0.042^{+0.091}_{-0.090}$	$0.083^{+0.012}_{-0.011}$
Free, Gaussian	$1.873^{+0.051}_{-0.056}$	$2.038^{+0.145}_{-0.168}$	$-0.947^{+0.348}_{-0.312}$	$-0.027^{+0.117}_{-0.125}$	$0.086^{+0.012}_{-0.011}$
Fixed 0	$1.810^{+0.046}_{-0.050}$	$1.912^{+0.166}_{-0.182}$	$-0.845^{+0.387}_{-0.343}$	$-0.074^{+0.132}_{-0.140}$	$0.089^{+0.012}_{-0.011}$
Fixed 0.18	$1.731^{+0.055}_{-0.057}$	$1.510^{+0.186}_{-0.172}$	$-0.310^{+0.399}_{-0.386}$	$-0.333^{+0.129}_{-0.107}$	$0.108^{+0.013}_{-0.012}$

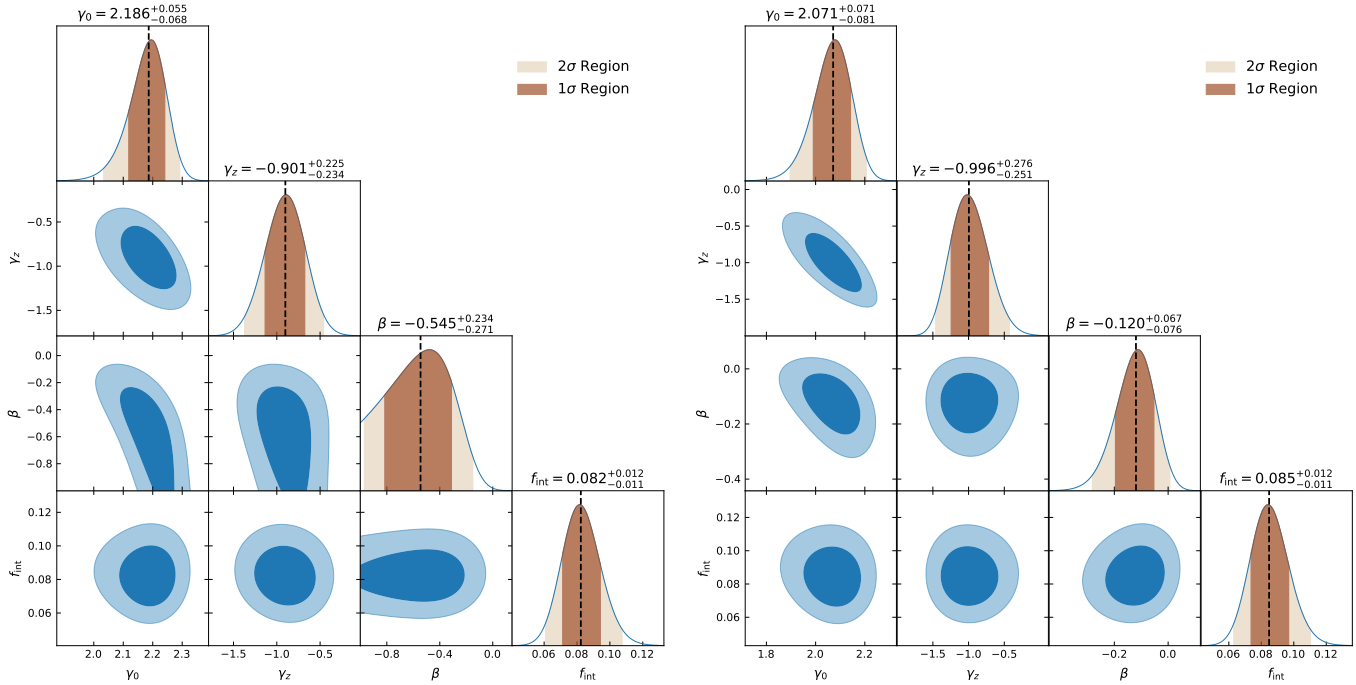


Fig. F.1. P2 individual-slope power-law tracer. Free beta with a uniform prior (left) and the truncated Gaussian prior (right). The marginal and contour conventions are those of Appendix F.

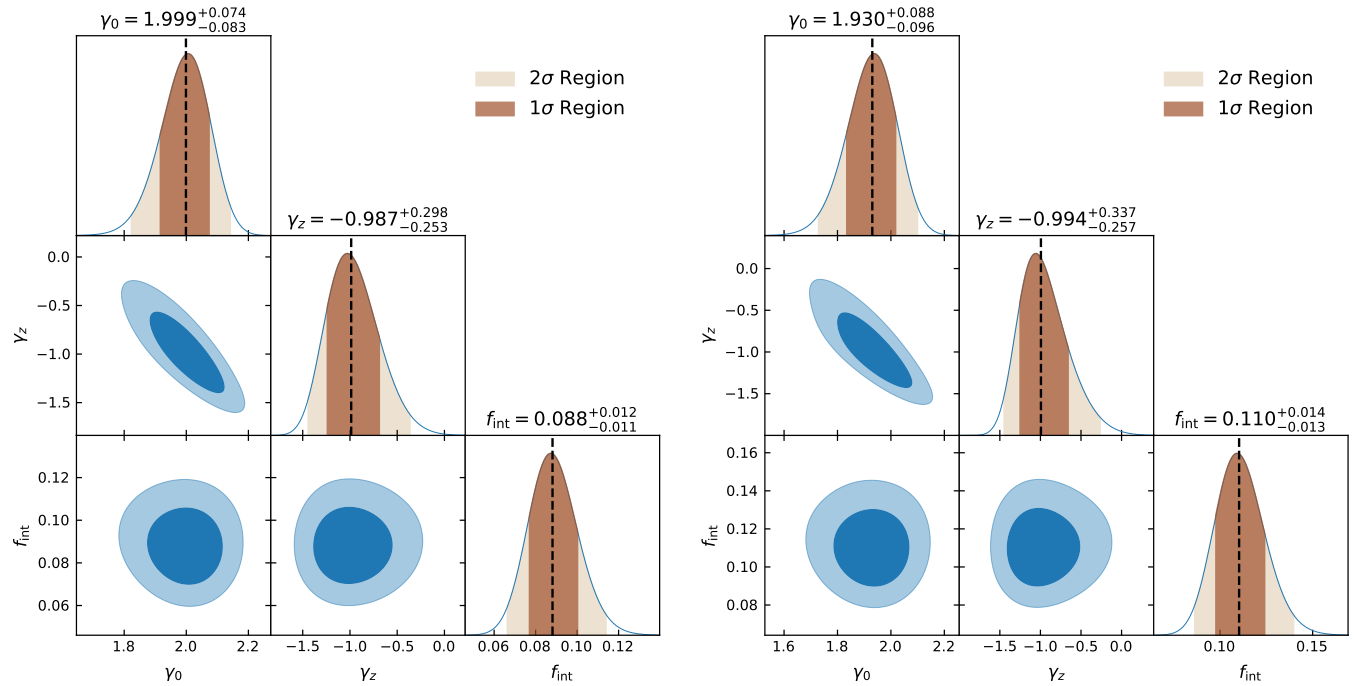


Fig. F.2. P2 individual-slope power-law tracer. Fixed $\beta = 0$ (left) and $\beta = 0.18$ (right). The marginal and contour conventions are those of Appendix F.

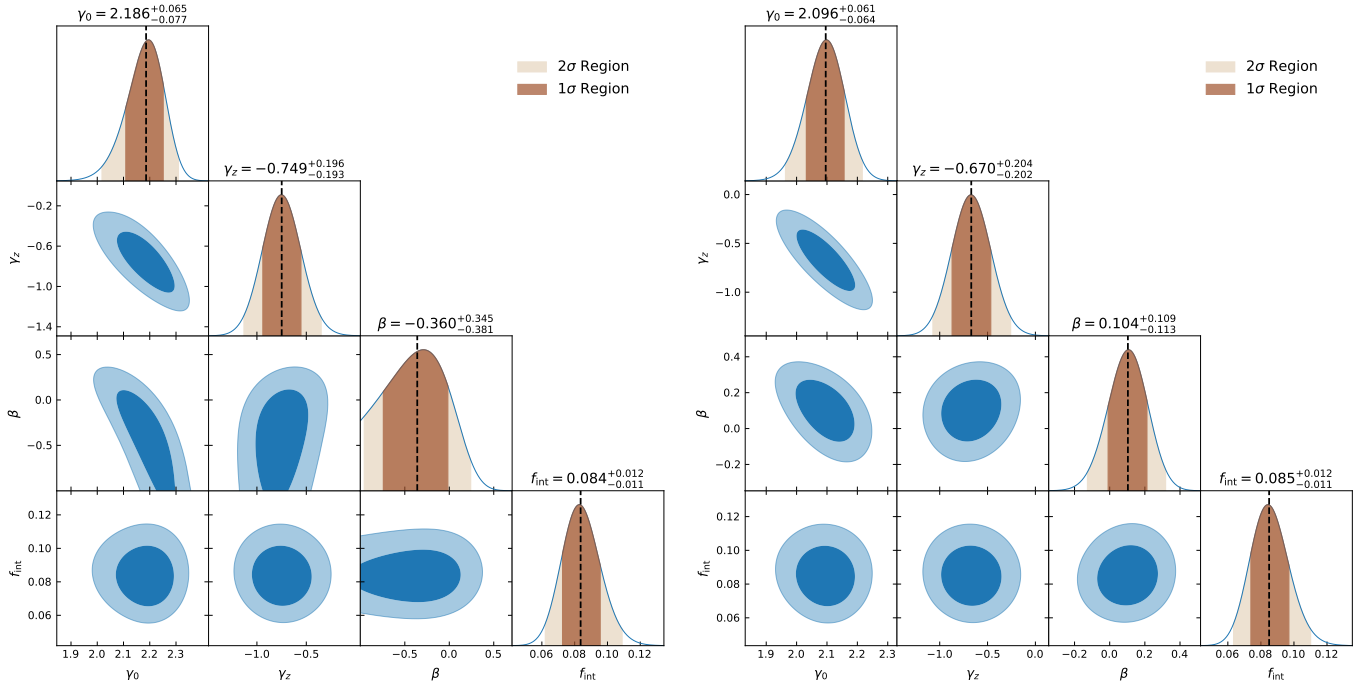


Fig. F.3. P2 Hernquist tracer. Free beta with a uniform prior (left) and the truncated Gaussian prior (right). The marginal and contour conventions are those of Appendix F.

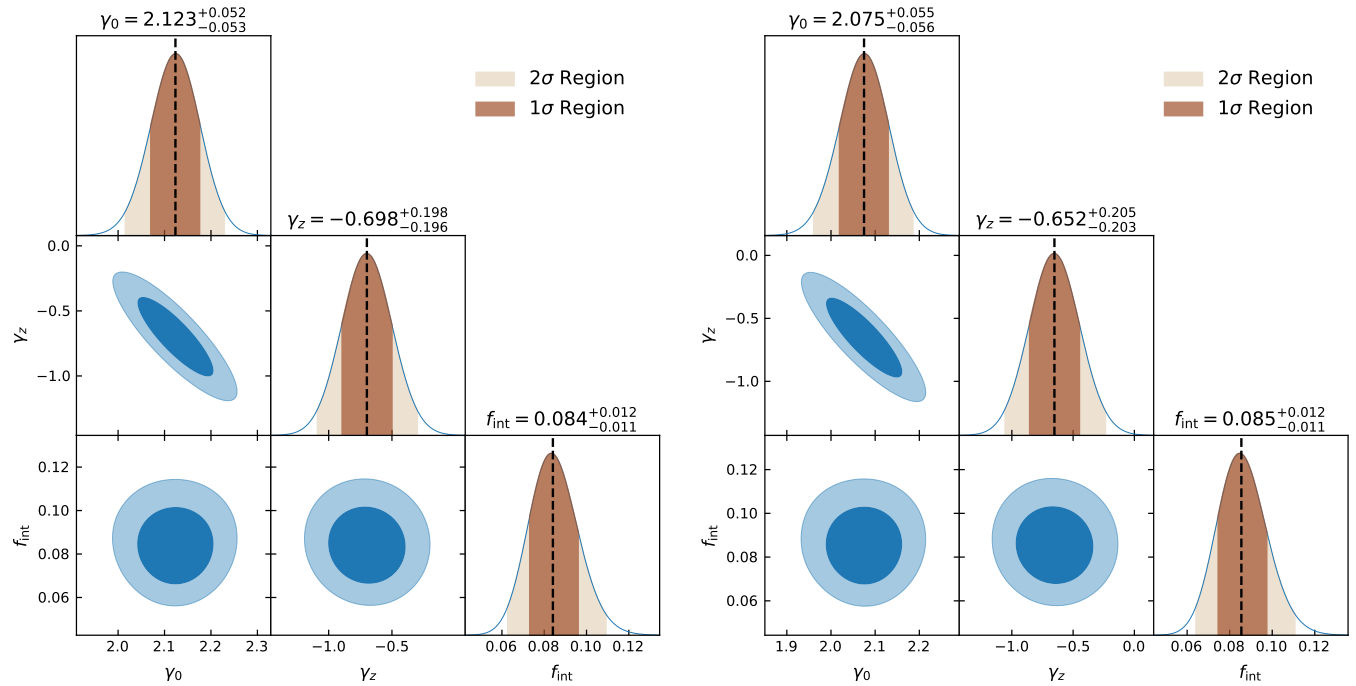


Fig. F.4. P2 Hernquist tracer. Fixed $\beta = 0$ (left) and $\beta = 0.18$ (right). The marginal and contour conventions are those of Appendix F.

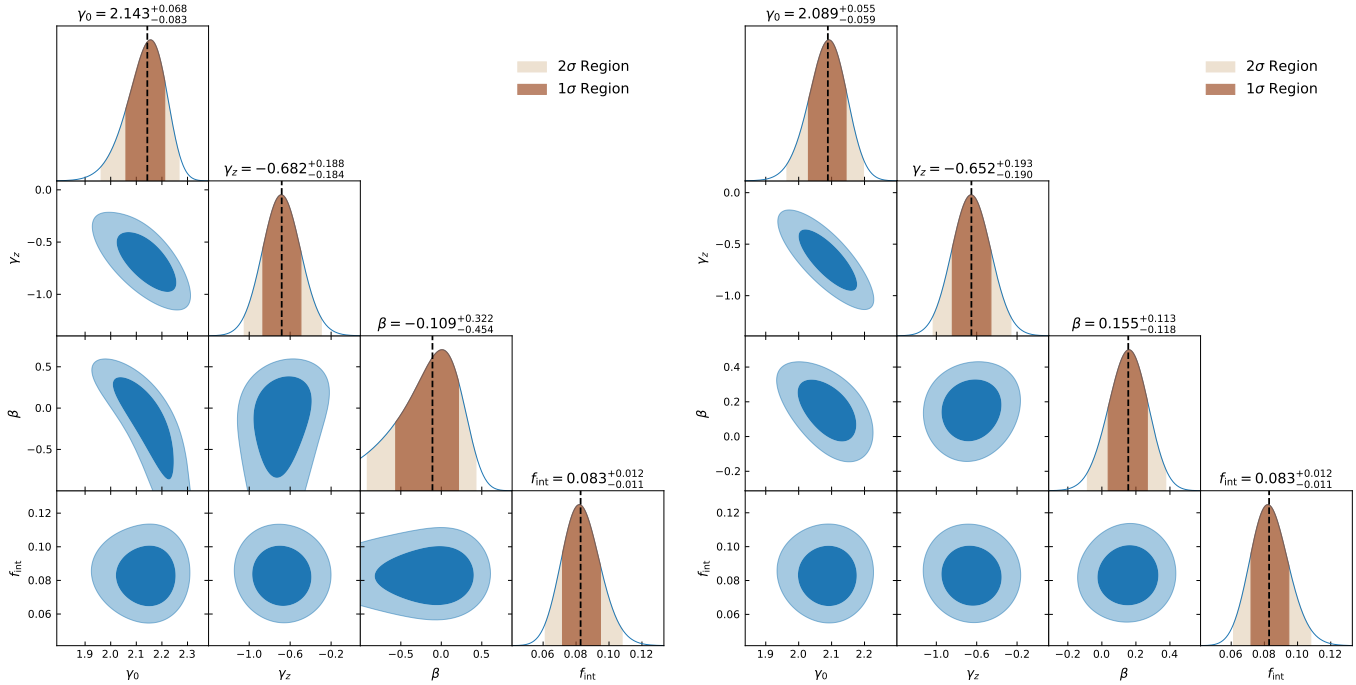


Fig. F.5. P2 Sérsic $n = 4$ tracer. Free beta with a uniform prior (left) and the truncated Gaussian prior (right). The marginal and contour conventions are those of Appendix F.

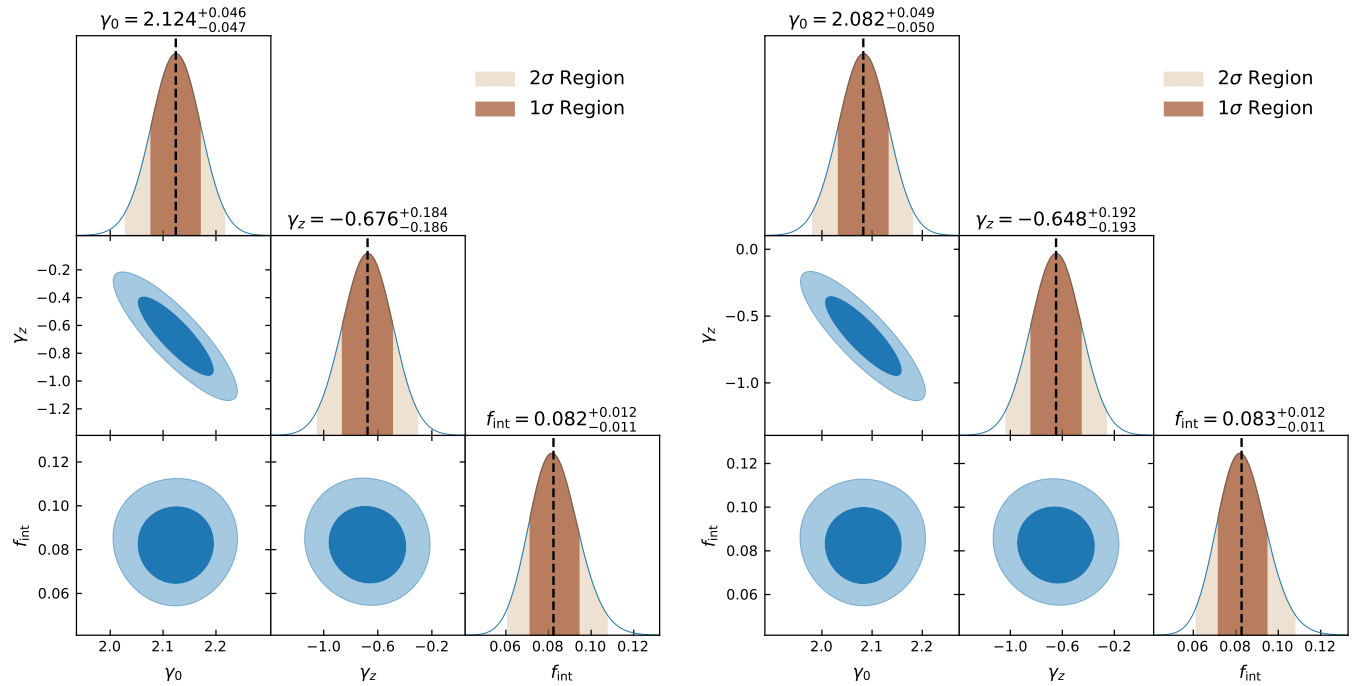


Fig. F.6. P2 Sérsic $n = 4$ tracer. Fixed $\beta = 0$ (left) and $\beta = 0.18$ (right). The marginal and contour conventions are those of Appendix F.

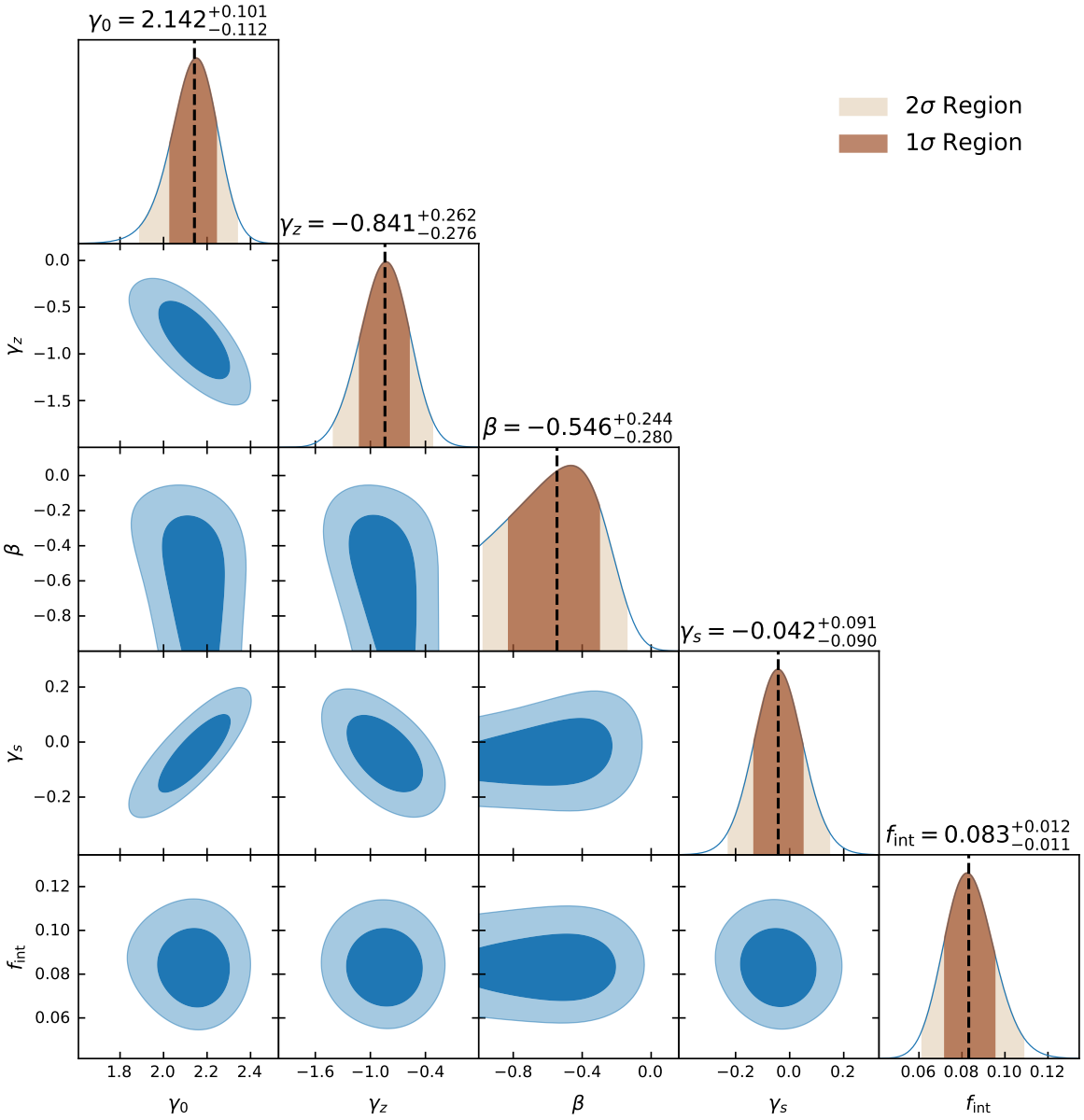


Fig. F.7. P3 individual-slope power-law posterior with a uniform beta prior. The additional structural coefficient γ_s multiplies the lensing-based predictor $\log_{10}\tilde{\Sigma}_L$.

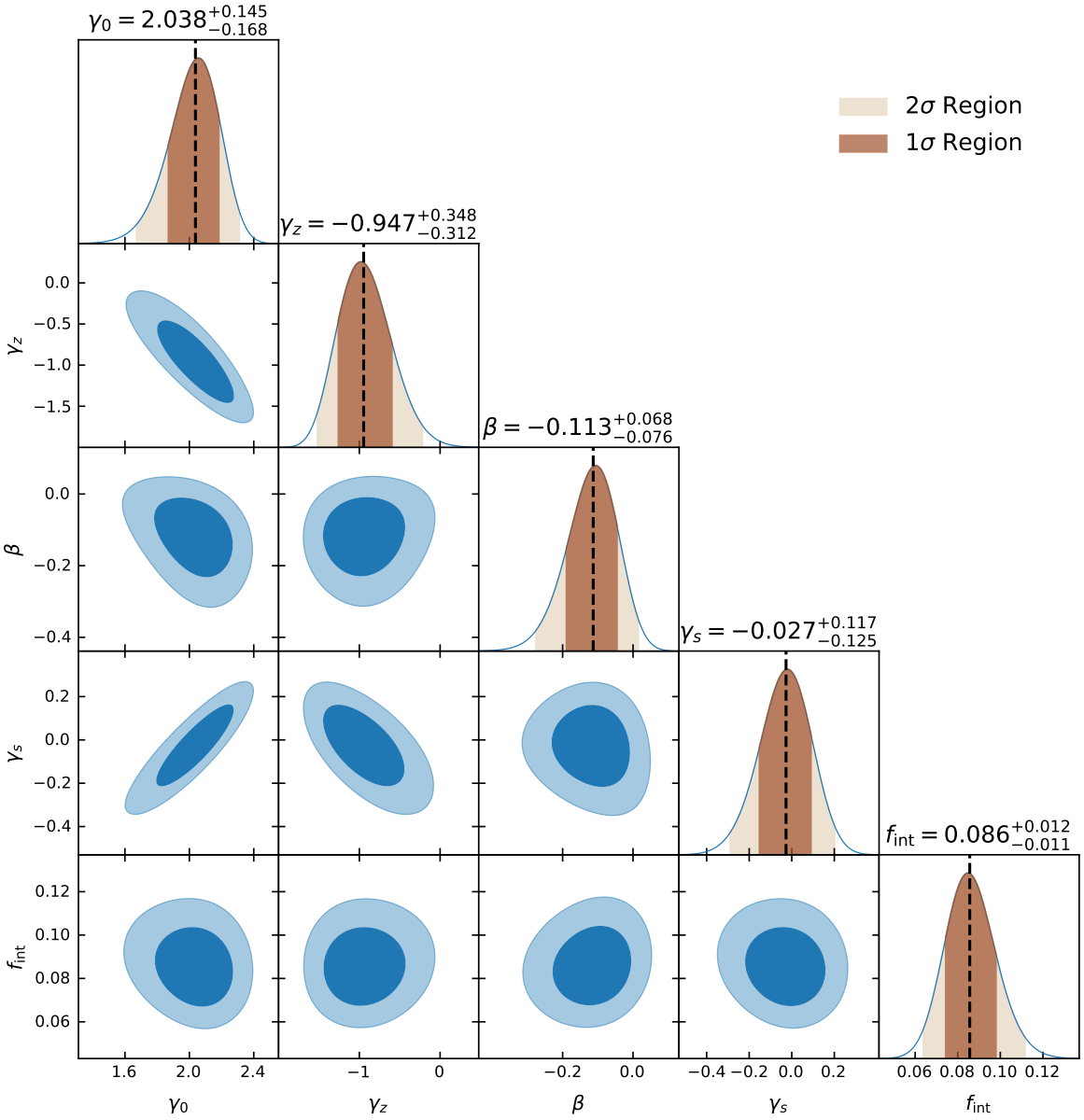


Fig. F.8. P3 individual-slope power-law posterior with the truncated Gaussian beta prior. The additional structural coefficient γ_s multiplies the lensing-based predictor $\log_{10}\widetilde{\Sigma}_L$.

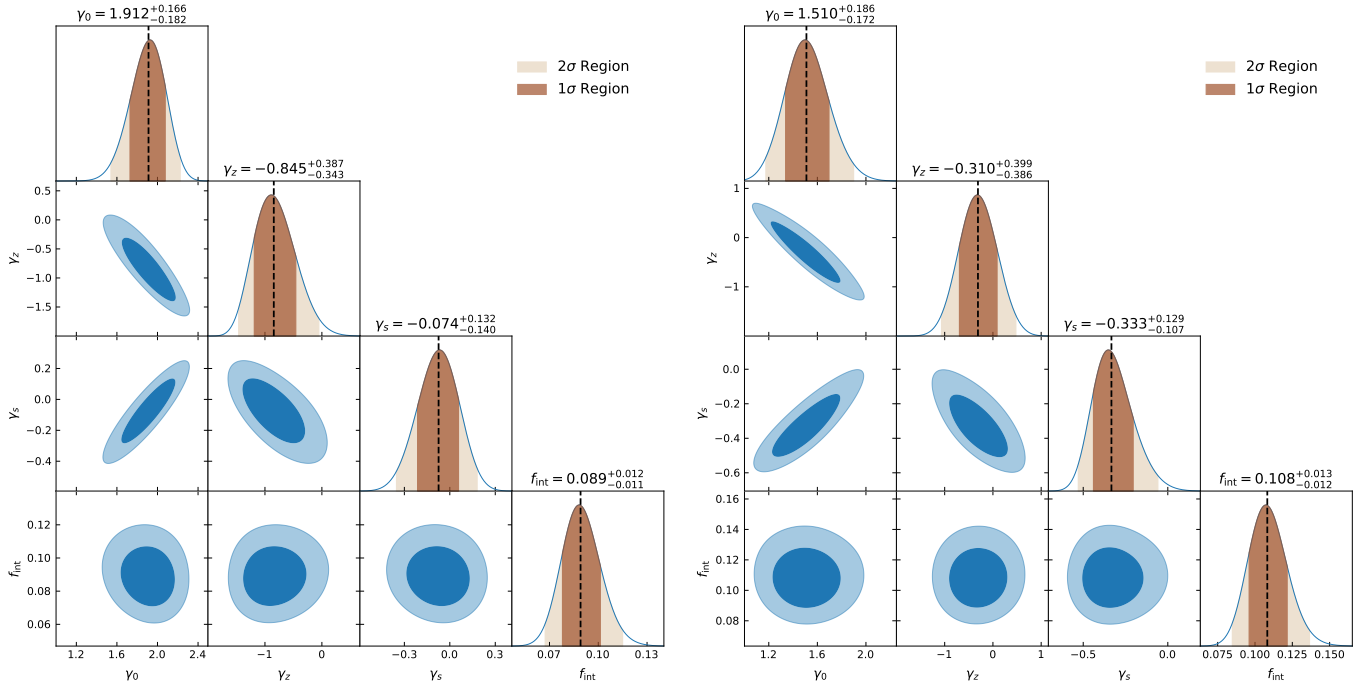


Fig. F.9. P3 individual-slope power-law posteriors with fixed $\beta = 0$ (left) and $\beta = 0.18$ (right). The structural predictor and other model assumptions are the same as in the free-beta P3 panels.